Solving Right Triangles: Elevation and Depression Answer Key

High School Geometry

Quick Answers

1. 5.30 m
2. 28.01 ft
3. 76.94 m
4. 53.1 degrees
5. 182.91 m
6. 160.40 ft
7. 44.99 m
8. (a) zip-line length = 48.46 m, (b) horizontal distance = 43.18 m, —

What is verified

7 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-SRT.C.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

2. *If they got 22.94: used sine instead of tangent, treating the 40 ft as the hypotenuse*
3. *If they got 8.12: multiplied by the tangent instead of dividing by it*
5. *If they got 78.73: multiplied by the sine instead of dividing by it*
6. *If they got 45.04: multiplied by the cosine instead of dividing by it*
7. *If they got 29.99: stopped at the height above the window and never added the window height*

1. A 6 m ladder makes an angle of 62° with level ground. How far up the wall does it reach, in metres?
Solution. The ladder is the hypotenuse and the wall height is the side opposite the 62° angle, so use sine:

$$\frac{\text{opposite}}{\text{hypotenuse}} = \sin 62^\circ \qquad \text{height} = 6 \sin 62^\circ = 5.2977\ldots$$

Rounded to two decimal places, $\boxed{5.30 \text{ m}}$

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2. A surveyor stands 40 ft from a flagpole; the angle of elevation to the top is 35°. Find the height above eye level, in feet.
Solution. The 40 ft run is adjacent to the angle and the height is opposite it, so use tangent:

$$\text{height} = 40\tan 35^\circ = 28.0083\ldots$$

Therefore $\boxed{28.01 \text{ ft}}$

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3. From the top of a 25 m lighthouse the angle of depression to a boat is 18°. Find the horizontal distance to the boat, in metres.
Solution. The angle of depression at the top equals the angle of elevation at the boat, so in the right triangle the 25 m height is opposite 18° and the horizontal distance is adjacent:

$$\tan 18^\circ = \frac{25}{d} \qquad d = \frac{25}{\tan 18^\circ} = 76.9421\ldots$$

Therefore $\boxed{76.94 \text{ m}}$

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4. A guy wire runs from an anchor 12 m from the base of a mast to a point 16 m up the mast. Find the angle of elevation of the wire, to the nearest tenth of a degree.
Solution. The height up the mast is opposite the angle and the ground distance is adjacent, so the tangent is known and the angle comes from the inverse tangent:

$$\tan x = \frac{16}{12} \qquad x = \tan^{-1}\left(\frac{16}{12}\right) = 53.1301\ldots^\circ$$

Therefore $\boxed{x = 53.1^\circ}$

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5. A drone hovers 120 m above ground; the angle of depression to a landing pad is 41°. Find the straight-line distance to the pad, in metres.
Solution. The 120 m altitude is opposite the 41° elevation angle at the pad, and the straight-line distance is the hypotenuse:

$$\sin 41^\circ = \frac{120}{c} \qquad c = \frac{120}{\sin 41^\circ} = 182.9104\ldots$$

Therefore $\boxed{182.91 \text{ m}}$

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6. A hiker stands 85 ft from a cliff base; the angle of elevation to the top is 58°. Find the length of the line of sight, in feet.
Solution. The 85 ft is adjacent to the 58° angle and the line of sight is the hypotenuse, so use cosine:

$$\cos 58^\circ = \frac{85}{c} \qquad c = \frac{85}{\cos 58^\circ} = 160.4018\ldots$$

Therefore $\boxed{160.40 \text{ ft}}$

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7. From a window 15 m above ground the angle of elevation to a tower top is 32°; the tower is 48 m away horizontally. Find the total height of the tower above the ground, in metres.
Solution. The right triangle sits above the window line: its adjacent side is 48 m and its opposite side is the part of the tower above the window.

$$\text{above window} = 48 \tan 32^\circ = 29.9937\ldots$$

The window itself is 15 m up, so the total height is $15 + 48 \tan 32^\circ = 44.9937\ldots$, giving $\boxed{44.99 \text{ m}}$

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8. A zip line runs from a platform 22 m above ground to a landing point; the angle of depression along the line is 27°. (a) Find the length of the zip line, in metres. (b) Find the horizontal distance from the base of the platform to the landing point, in metres. (c) Explain why the angle of depression at the platform equals the angle of elevation at the landing point.
Solution (a). The 22 m drop is opposite the 27° angle at the landing point and the zip line is the hypotenuse:

$$c = \frac{22}{\sin 27^\circ} = 48.4592\ldots$$

Therefore $\boxed{48.46 \text{ m}}$
Solution (b). The horizontal distance is adjacent to the same angle:

$$d = \frac{22}{\tan 27^\circ} = 43.1774\ldots$$

Therefore $\boxed{43.18 \text{ m}}$
Solution (c) — instructor-judged. A complete response says the horizontal line at the platform and the ground are parallel, and the zip line is a transversal cutting both, so the angle of depression at the top and the angle of elevation at the bottom are alternate interior angles and therefore equal. Award full credit only when the parallel lines and the alternate-interior angle relationship both appear; simply asserting that the two angles are the same is not a justification.